

#HW 11

● Graded

Student

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Total Points

95 / 100 pts

Question 1

(no title)

30 / 30 pts

✓ - 0 pts Correct

Question 2

(no title)

9 / 10 pts

- 0 pts Correct

- 1 pt for part b, the process can be possible as entropy transfer can be less than 0.

- 1 pt c, will be impossible. To make the entropy change 0 when the entropy transfer is more than 0, you need to have a negative entropy production, which is not possible.

✓ - 1 pt d will be indeterminate because if entropy change is less than the entropy transfer then energy production can become negative which is not possible. Only when the entropy change is greater than or equal to the entropy transfer, the entropy production will be greater than or equal to 0 which makes the process possible.

- 1 pt g will be indeterminate too.

- 1 pt e will be possible.

- 5 pts It has not been explained whether the process will be possible or impossible or indeterminate.

- 1 pt f will be impossible since entropy production should always be positive.

- 1 pt a will be possible because in absence of entropy transfer, the entropy production must be positive to make entropy change greater than 0.

Question 3

(no title)

15 / 15 pts

✓ - 0 pts Correct

Question 4

(no title)

11 / 15 pts

✓ - 0 pts Correct

- 2 pts Part c will be irreversible since, Q_h will be 1000 and as a result efficiency will be 60 percent.

✓ - 2 pts Part a will be irreversible since efficiency will be 0.5, which is less than $2/3$.

✓ - 2 pts b will be reversible since efficiency will be $2/3$. The work will be 600 kJ.

- 5 pts You have to mention which process is reversible, which is irreversible, and which is impossible.

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts For efficiency calculation, temperatures should always be converted to Kelvin or Rankine.

- 2 pts In part b, efficiency will be around 0.6. Answers will be different

- 1 pt the answer of c is not clear.

- 2 pts In part d T_h is 1000 and you need to find T_c .

Question 6

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 1 pt The efficiency calculation is wrong.

Question assigned to the following page: [1](#)

6.17.3

Wednesday, November 12, 2025

2:55 PM

(a) Classify the following processes of a closed system as *possible*, *impossible*, or *indeterminate*.

	Entropy Change	Entropy Transfer	Entropy Production
(a)	> 0	0	> 0
(b)	< 0	< 0	> 0
(c)	0	> 0	< 0
(d)	> 0	> 0	≥ 0
(e)	0	< 0	> 0
(f)	> 0	> 0	< 0
(g)	< 0	< 0	≤ 0

a) possible

b) possible

c) impossible

d) possible

Question assigned to the following page: [1](#)

- e) possible
- f) impossible
- g) indeterminate

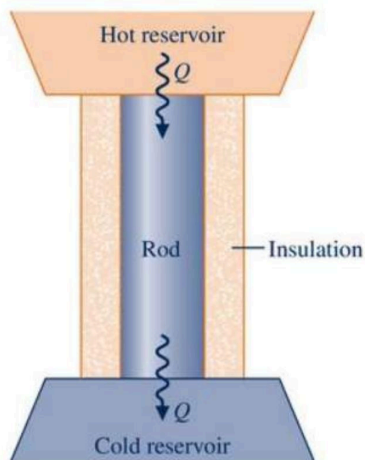
Question assigned to the following page: [2](#)

6.17.9

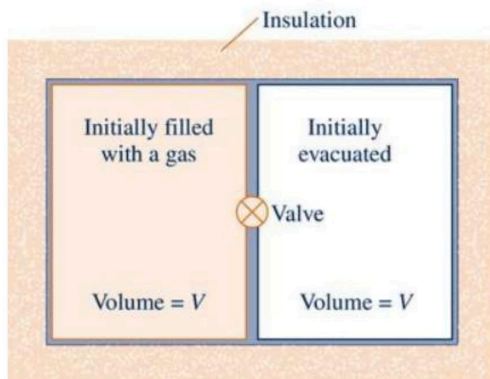
Wednesday, November 12, 2025 2:55 PM

Use the Kelvin-Planck statement of the second law to show that the specified process is irreversible.

- (a) As shown in the figure below, a hot thermal reservoir is separated from a cold thermal reservoir by a cylindrical rod insulated on its lateral surface. Energy transfer by conduction between the two reservoirs takes place through the rod, which remains at steady state.



- (b) As shown in the figure below, a rigid insulated tank is divided into halves by a partition. On one side of the partition is a gas. The other side is initially evacuated. A valve in the partition is opened and the gas expands to fill the entire volume.



- A) Because energy can't be made to go from the cold reservoir to the hot reservoir without putting in work, the process is irreversible.
- B) Because the gas would not be able to be made to go back to its original side of the tank without putting in work, the process is irreversible.

Question assigned to the following page: [3](#)

6.17.14

Wednesday, November 12, 2025 2:55 PM

A power cycle receives Q_H by heat transfer from a hot reservoir at $T_H = 1200$ K and rejects energy Q_C by heat transfer to a cold reservoir at $T_C = 400$ K. For each of the following cases, determine whether the cycle operates *reversibly*, operates *irreversibly*, or is *impossible*.

(a) $Q_H = 900$ kJ, $W_{\text{cycle}} = 450$ kJ

(b) $Q_H = 900$ kJ, $Q_C = 300$ kJ

(c) $W_{\text{cycle}} = 600$ kJ, $Q_C = 400$ kJ

(d) $\eta = 75\%$

A) Reversible

B) Irreversible

C) Irreversible

D) Impossible

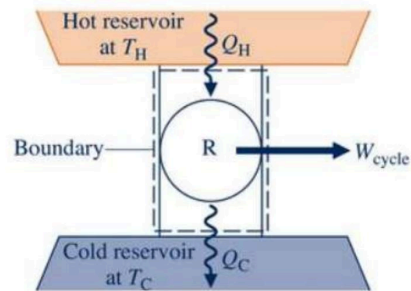
$$\text{Efficiency} = 2/3$$

Question assigned to the following page: [4](#)

6.17.17

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As shown in the figure below, a reversible power cycle receives energy Q_H by heat transfer from a hot reservoir at T_H and rejects energy Q_C by heat transfer to a cold reservoir at T_C .



- (a) If $T_H = 1600 \text{ K}$ and $T_C = 400 \text{ K}$, what is the thermal efficiency?
- (b) If $T_H = 500^\circ\text{C}$, $T_C = 20^\circ\text{C}$, and $W_{\text{cycle}} = 1000 \text{ kJ}$, what are Q_H and Q_C , each in kJ?
- (c) If $\eta = 60\%$ and $T_C = 40^\circ\text{F}$, what is T_H , in $^\circ\text{F}$?
- (d) If $\eta = 40\%$ and $T_H = 727^\circ\text{C}$, what is T_C , in $^\circ\text{C}$?

$$\eta = \frac{W_{\text{cycle}}}{Q_H} = 1 - \frac{Q_C}{Q_H}$$

$$a) \quad \eta = 1 - \frac{400 \text{ K}}{1600 \text{ K}} = \boxed{0.75}$$

$$b) \quad \eta = 1 - \frac{(20 + 273) \text{ K}}{(500 + 273) \text{ K}} = 0.62$$

$$0.62 = \frac{1000 \text{ kJ}}{Q_H} \quad \boxed{Q_H = 1610.42 \text{ kJ}}$$

Question assigned to the following page: [4](#)

$$0.62 = 1 - \frac{Q_c}{1610.42 \text{ kJ}}$$

$$Q_c = 610.42 \text{ kJ}$$

$$c) \quad 0.6 = 1 - \frac{(40 + 460)^\circ \text{R}}{(T_H + 460)^\circ \text{R}}$$

$$T_H = 790^\circ \text{F}$$

$$d) \quad 0.4 = 1 - \frac{(T_c + 273) \text{ K}}{(727 + 273) \text{ K}}$$

$$T_c = 327^\circ \text{C}$$

Questions assigned to the following page: [5](#) and [6](#)

6.17.19

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- (a) At a particular location, magma exists several kilometers below Earth's surface at a temperature of 1100°C , while the average temperature of the atmosphere at the surface is 15°C . An inventor claims to have devised a power cycle operating between these temperatures having a thermal efficiency of 79%. Investigate this claim.

$$T_H = 1100^{\circ}\text{C} = 1373 \text{ K}$$

$$T_C = 15^{\circ}\text{C} = 288 \text{ K}$$

$$\eta = 1 - \frac{288 \text{ K}}{1373 \text{ K}} = \underline{0.79}$$

It would appear that this claim is
true.